Matthis Lahargoue

Machine Learning · Computer Vision · Scientific Computing

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Engineering student on an ENSISA/UQAC double-degree pathway, currently a computational imaging research intern at ISL. Focused on machine learning, inverse problems, reproducible experimentation and scientific computing. Interested in ML and vision opportunities, especially in Switzerland.

RESEARCH & PROFESSIONAL EXPERIENCE

Research Intern - Computational Imaging / NLOS

French-German Research Institute of Saint-Louis (ISL) | July 2026 - present

Research on learning-based non-line-of-sight reconstruction. Work connects transient-light simulation, data preparation, model training and quantitative evaluation. Tools: PyTorch, Mitsuba 3, Dr.Jit, CUDA, Linux and HPC. Public description only; institutional research details remain confidential.

Computer Vision / Applied ML Project

MAGYAR SA | June 2025

Developed an offline CNN prototype for industrial tank image classification in a two-person team. Curated approximately 1,000 proprietary images. Addressed class imbalance, visual similarity and variable acquisition conditions with data preparation, transfer learning and class-level evaluation.

EDUCATION

Engineering Degree - Computer Science & Network Engineering

ENSISA, Université de Haute-Alsace, France | In progress; diploma expected May 2027

Master's in Computer Science - Artificial Intelligence

Université du Québec à Chicoutimi (UQAC), Canada | 2025-2026 studies; degree pending

SELECTED PROJECTS

CNN Stock Prediction - Reproducibility Study

Public reproduction study using chronological evaluation, train-only normalization and explicit naive baselines. The investigated CNNs did not improve on the reported always-BUY baseline. Findings are specific to the tested setup.

View public repository

Basel Tram Air - Hack am Rhein Warm-up

Interactive exploration of historical PM2.5 measurements along Basel tram corridors. Hourly aggregation, 262 spatial cells and a public map with slider/timelapse. Historical observations, not live measurements.

View public repository

Large-Scale Sentiment Analysis - UQAC

Graduate project with PySpark, MinHash LSH and approximate k-nearest neighbors on Calcul Québec HPC infrastructure, studying representations and scaling on Sentiment140.

TECHNICAL TOOLKIT

Python · PyTorch · Scikit-learn · Computer vision · Inverse problems · Mitsuba 3 · Dr.Jit · CUDA · NumPy · Pandas · PySpark · Git · Linux · HPC / Slurm

French: native · English: B2 (Linguaskill) · German: B1 (Goethe-Test PRO)